Training components: trailing mean of 50 updates from the same source (100 mixed updates)

Learned-noise control
Learned noise + variogram
Shared deep + wide control
Shared deep + wide + variogram

OM4 initializer: base fair CRPS × 0.1

OM4 native forecast loss × 0.5

Observation total loss

Added variogram contribution × 0.1

Variogram fair score, unweighted

Finite-ensemble correction, unweighted

Normalized training loss

Total mixed optimizer updates